

SAQIB SHAHZAD QAZI

Senior AI/ML Engineer | GenAI & Agentic Systems

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SKILLS OVERVIEW

- **Agentic AI & Orchestration:** Multi-agent system design with CrewAI, LangGraph, AutoGen, and ReAct-style reasoning; tool use, memory, human-in-the-loop escalation, and fallback strategies across autonomous pipelines.
- **LLM Engineering:** LangChain, GPT-4o, Claude family models, open-source LLMs; prompt engineering, chaining, RAG over vector stores (Pinecone, FAISS, Chroma), and agent-based pipelines that translate natural language into safe, structured actions.
- **Enterprise AI Solutions:** AI-powered assistants for GRC, banking compliance, procurement, and document intelligence with guardrails, confidence scoring, and validation layers; multi-agent orchestration and fallback strategies.
- **Voice AI:** Vapi.ai, Azure STT/TTS, Groq; real-time telephony pipelines, LLM-driven conversation logic, HIPAA-aware audio design with no audio persistence.
- **Python & API Stack:** FastAPI, Flask, Dash; pandas, NumPy, scikit-learn, TensorFlow, PyTorch, NLP tooling; production APIs, async services, and inference endpoints.
- **Data Engineering:** Production-grade ETL pipelines with Python and Airflow; automated ingestion from APIs, databases, and documents; OCR-driven workflows (Tesseract) with downstream LLM reasoning.
- **Cloud & DevOps:** AWS (Lambda, API Gateway), GCP, Azure; Docker, Kubernetes, CI/CD pipelines; deploying and scaling production AI services.
- **AI Safety & Reliability:** Confidence scoring, hallucination mitigation, RBAC, tenant isolation, audit-ready logging, and human-in-the-loop escalation across enterprise deployments.
- **Analytics & Visualization:** Power BI, Dash, Plotly; decision-critical dashboards; strong SQL across relational databases; Snowflake and enterprise data systems.
- **Work Model:** Fully remote (US); open to full-time, contract-to-hire, and trial-based engagements; experienced in async, agile delivery.

PROFESSIONAL EXPERIENCE

Senior AI/ML Engineer — TriNet

Jul 2025 – May 2026

- Architected and delivered AI-powered GRC and procurement assistants, enabling auditors, risk managers, and procurement teams to query policies, controls, vendors, and risks via natural language interfaces.
- Led a banking-sector engagement with a Tier-1 North American bank (under NDA); built a compliance intelligence layer over their regulatory schema, automating control mapping, audit evidence retrieval, and regulatory reporting with LLM-powered RAG pipelines.
- Designed safe multi-table agentic query pipelines across complex enterprise schemas with full role-based access control and tenant isolation, cutting audit preparation time by 60–70%.
- Integrated LLM agents with backend APIs for risk summaries, compliance dashboards, and policy gap analysis; implemented guardrails and confidence scoring to meet enterprise and regulatory standards.
- Collaborated cross-functionally with product, compliance, and engineering teams to align AI outputs with real enterprise workflows and audit requirements.

AI/ML Engineer (GenAI & Enterprise Systems) — CureMD RCM

Jul 2021 – Jun 2025

- Architected a multi-agent GenAI platform (planner, tool-use, and verifier stages) that automated end-to-end employee onboarding and benefits-resolution workflows, cutting manual case-handling time by 65% across HR operations.
- Built a production RAG system over 800K+ HR policy, payroll, and compliance documents with hybrid retrieval and re-ranking, improving answer relevance by 38% and reducing hallucinated responses by 50%.
- Productionized LLM inference services handling 1.5M+ requests/day at p95 under 300ms using async FastAPI, request batching, and Kubernetes autoscaling, while lowering compute cost per query by 40%.
- Established the team's AI safety and evaluation layer, including confidence scoring, hallucination detection, RBAC, and an LLM-as-judge eval suite across 12+ quality dimensions, enabling regression-gated releases into compliance-sensitive HR environments.
- Led and mentored a 4-engineer ML pod, setting standards for prompt versioning, guardrails, and MLOps (CI/CD, drift monitoring, automated rollback) that reduced time-to-production for new AI features by 45%.

Data Scientist & ML Engineer (Leading GenAI) — Mind Maze

Dec 2018 – Jun 2021

- Led a team of 5 analysts to build enterprise-wide KPI dashboards used by senior leadership for weekly decision-making across sales, product, and marketing.
- Analyzed 10M+ data rows across marketing, finance, and operations; delivered strategic insights that contributed to a 15% revenue lift and quarterly C-level planning.
- Initiated and delivered an LLM-powered business intelligence reporting engine using LangChain; agents summarize KPI data, surface anomalies, and auto-generate executive briefings.
- Designed and deployed an OCR-powered text extraction and auto-summarization pipeline using Tesseract and LangChain, improving document processing speed by 50%.

- Automated reporting pipelines with Python and Excel macros; reduced manual reporting time by 40% and delivered executive-ready reports in near real-time.

KEY PROJECTS

Agentic Procurement Intelligence Platform

CrewAI · LangChain · GPT-4o · FastAPI · Pinecone · AWS

- Architected a fully autonomous multi-agent procurement workflow (Intake, Vendor Intelligence, Approval Orchestrator, Audit Trail), enabling natural-language PO resolution with zero manual triage.
- Built a Pinecone vector store over 50K+ procurement docs for real-time clause retrieval and contract risk flagging; integrated live ERP APIs with guardrails for spend-threshold enforcement and human-in-the-loop escalation for high-value decisions.
- Reduced procurement cycle time by 55% in pilot; deployed on AWS Lambda + API Gateway with a FastAPI backend and audit-ready logging, fully compatible with enterprise GRC and SOX compliance requirements.

Enterprise GRC & Compliance Intelligence

LangChain · GPT-4o · RAG · Pinecone · FastAPI

- Led end-to-end development of an AI-driven GRC platform enabling auditors and compliance teams to query controls, risks, and policies via natural language across complex enterprise schemas, using LLM-powered RAG pipelines with validation layers and confidence scoring.
- Banking-sector deployment with a Tier-1 North American bank (under NDA) automated regulatory control mapping, compliance gap analysis, and audit evidence generation, cutting manual audit prep time by 60–70%.
- Built role-based access controls, tenant isolation, and explainability traces across multi-client deployments; improved audit decision turnaround by 40%.

Voice AI Patient Registration System

Vapi.ai · GPT-4o-mini · FastAPI · PostgreSQL · Railway

- Built a production-ready voice AI agent answering real U.S. phone calls, conducting natural LLM-driven conversations to collect patient demographics, validate input in real time, and persist confirmed records to PostgreSQL with duplicate detection and a soft-delete audit trail.
- Architected a 4-layer pipeline: Vapi.ai (telephony + STT/TTS), GPT-4o-mini (conversation logic), FastAPI (validation + business logic), PostgreSQL; deployed on Railway with a live REST API, full CRUD, and a web dashboard for record management.
- Implemented UUID primary keys for horizontal scalability, async endpoints for concurrent call handling, and HIPAA-aware data design with no audio persistence.

OMANI Therapist Voice

Azure STT/TTS · GPT-4o · Claude Opus 4 · Groq · Flask · Docker · Hugging Face

- Engineered a real-time, voice-only mental-health chatbot delivering culturally sensitive therapy in Omani Arabic, achieving sub-20s end-to-end latency (avg. 12–15s) via a pipeline of Azure STT, dual-model emotion detection, safety screening, GPT-4o/Claude Opus 4 response, and Azure TTS.
- Implemented a fine-tuned Hugging Face emotion model on annotated Omani Arabic phrases with CBT-based therapy response generation, crisis-keyword safety screening with escalation protocols, and cultural embeddings for authentic dialect adaptation.
- Deployed on Docker at zero monthly cost using Azure Free Tier (STT/TTS) and Groq free-tier LLM inference; designed with HIPAA-aware logging, no audio persistence beyond immediate processing, and Kubernetes-compatible architecture for production scaling.

Real-Time Healthcare Readmission Risk Prediction (ML + GenAI)

XGBoost · Dash · Flask · Plotly · scikit-learn

- Engineered a predictive XGBoost model on a 1M-row dataset with 5-fold cross-validation; automated risk scoring for 200K test patients, reducing readmissions by 15% and saving \$3.5M annually.
- Built a real-time Dash dashboard with a Flask API enabling 30-second updates across 6 visualizations for 1K daily patient queries, cutting intervention delays by 30% and saving \$1.2M in operational costs.
- Identified a 15% winter readmission spike in the 61–80 age group across 600K records, optimizing resource allocation to reduce peak-season readmissions by 10% and saving an additional \$2M.

EDUCATION

- M.S. Computer Science — Superior University, Lahore (Sep 2015 – Aug 2017)
- B.S. Computer Science — FAST University, Lahore (Jul 2011 – Jun 2015)

REFERENCES

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